Bio-inspired Design Proposal

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Design Theme

Locomotion

Designs that directly effect movement or interact/ contact direct movement

Focused Object

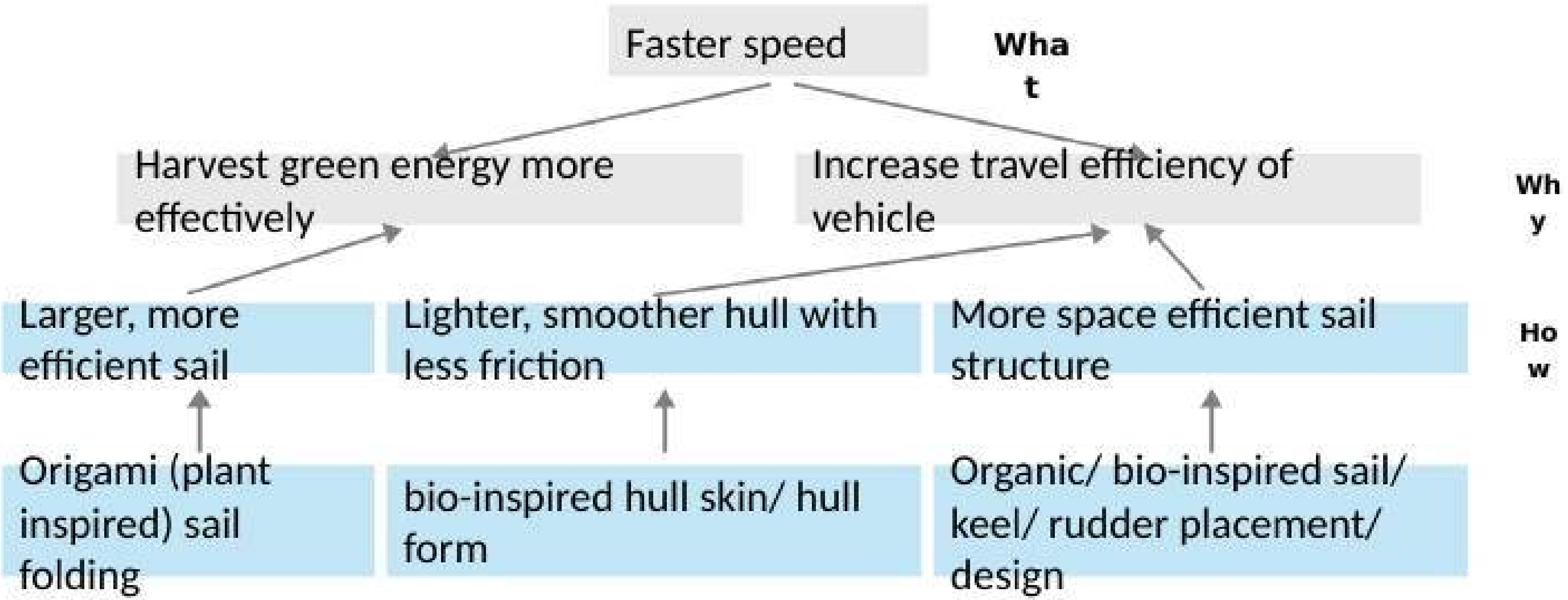
Sailboat

A boat that is propelled mainly by sails using the power of the wind, though many also carry small engines or oars for maneuvering.

What effects the mobility of a sailboat?

Component Decomposition of Human Design: Sailboat

Functional Decomposition of Human Design: Sailboat



Selected Organisms


Flying Fish


Duck


Mimosa Plant


Bats

Functional Decomposition of Organisms: Flying Fish Taxi

Flying fish taxiing in a manner comparable to sailing

Gliding height is an order of full span of pectoral wings

Reduction of drag and increase of lift-to-drag ratio by flying close to sea surface

Flying fish emerge at angles around 30 to 35 degrees

Largest lift coefficient at that angle

A stiff plate by the radials and well regulated muscles support the large pectoral fins

Increased stability and strength of fins during flight

Caudal complex has fused bones to provide force and stability

Flying fish gliding performance is similar to bird wings, but induced drag by wing-tip vortices is one of the main drag components

Low energy escape maneuver that can be controlled by flexor-extensor and adductor muscles

Citations: Black = 1 / Blue = 2

Functional Decomposition of Organisms: Duck's

Ducks use their webbed feet for air and water support

They have webbed feat that can expand and contract

They have highly flexible toes that are controlled by muscles

They paddle their feet in water to propel themselves forward

Using their flexible toes, they orient the webbed feet to best control their speed in water

They use drag from their webs to push water backwards

They use their feet as a rudder in air to help maneuver

They close their webbed feet and orient it underneath themselves. They essentially "point their toes" in either direction to direct themselves in flight

Citation 3

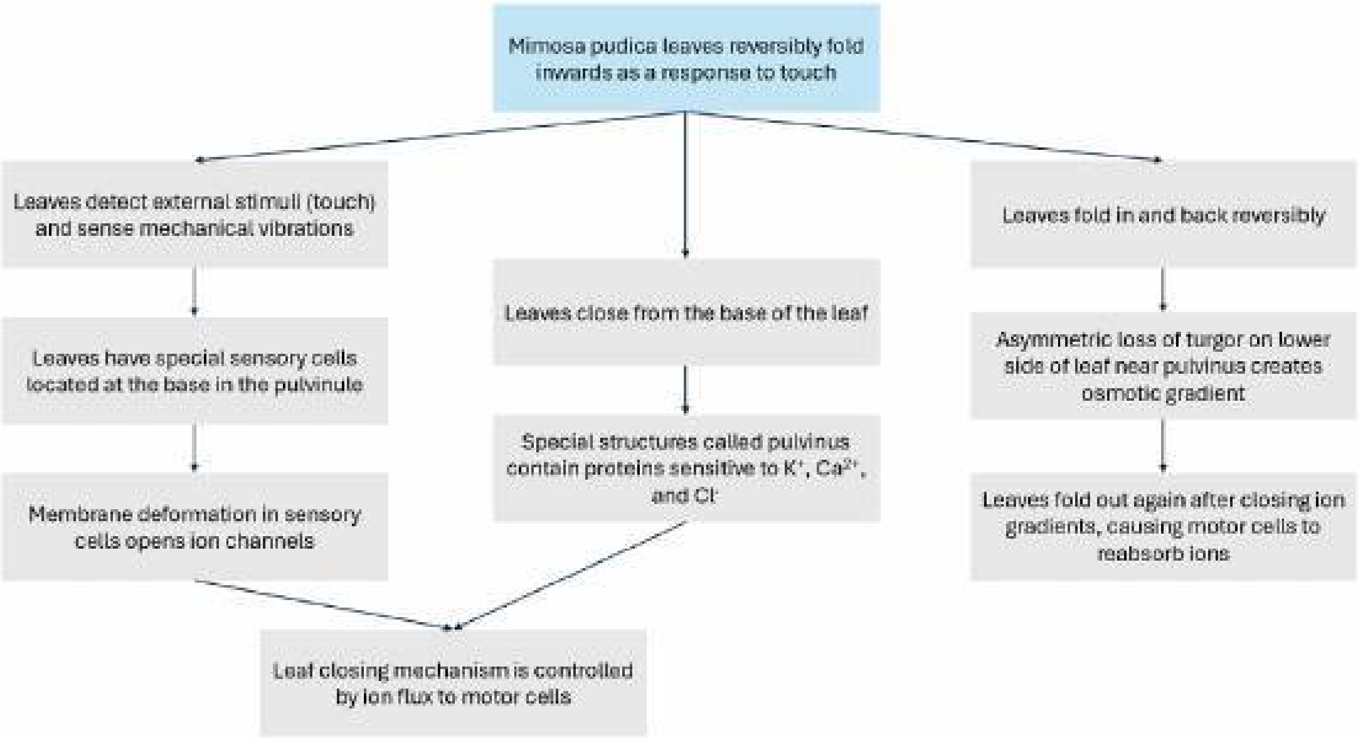
Functional Decomposition of Organisms: Bat Wing Modular Adjustment

- Bats have specialized maneuverability and agility during flight
  - Less stiff bones and aeroelastic membrane
    - Creates more deformation - promotes bending and shape changing of wing during flight to adjust steering and agility
  - "Hand" skeleton shape allows for multiple changes in wing properties
    - Provides structure and also ability to fold into compact space
  - Muscles adjust camber and stiffness to modulate lift and thrust throughout the wingbeat cycle
    - Creates leading edge vortices = more lift

FOR THE SAIL:

- Actuators on the supporting frame could subtly change the sail's curvature in real-time
- Sail can fold open while still having structure
- Aeroelastic material can help with deformation for thrust
- Better steering and agility of boat
- Light bones = lighter frame for reducing weight on the boat and increasing capture of energy from wind

Functional Decomposition of Organisms: *Mimosa pudica* Foldable Leaves



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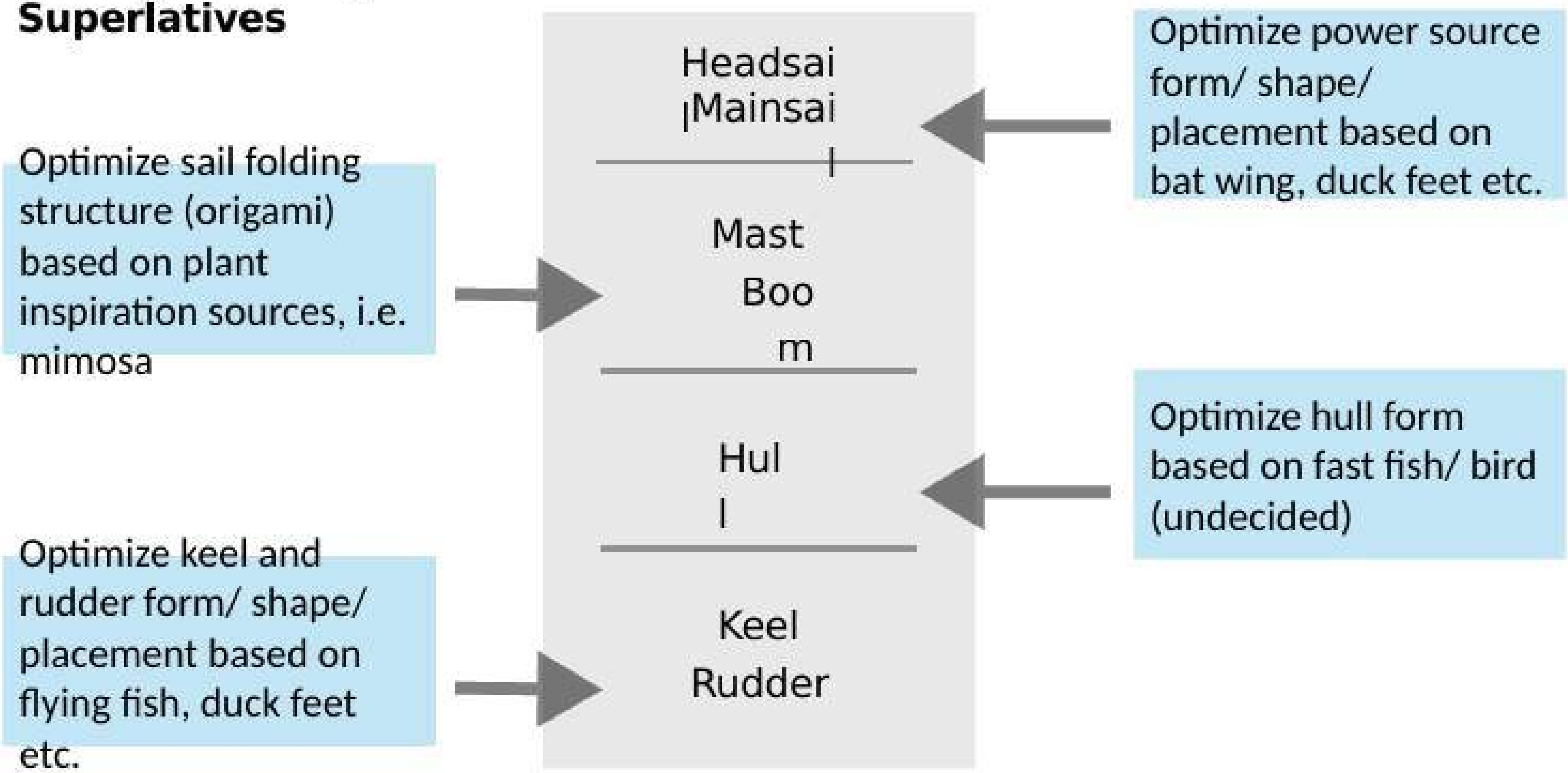
Translated Components and Superlatives



Translated Components and Superlatives

Drag Control: **Flying fish adjust wingspan for lift; ducks vary toe spread to change drag.**
Multi-Function Use: **Ducks' webbed feet speed up or slow down; flying fish use wings for flight & swimming.**
Stability vs Flexibility: **Bats adjust wing bones for agility → inspire foldable mast; mimosa folds & reopens reliably → precise mast folding.**
Adaptive Response: **Mimosa automatically opens/closes leaves to match conditions.**

Next Steps:

Research sailboat travel → identify least efficient design aspects
Combine animal strategies to improve mast & sail (synergy of benefits)
Develop testing system: • Wind/aero → sail efficiency & maneuverability • Foldability → mast performance
Build prototype and run trials
Analyze results → compare with current rigs'
Look into other potential source of bio-inspirations
Use data to explore potential sailing improvements

Bibliography

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